



Natural Language Processing (NLP)

Intro NLP

Equipping You with Research Depth and
Industry Skills

By:

Dr. Zohair Ahmed



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Recommended Books

Speech and Language Processing

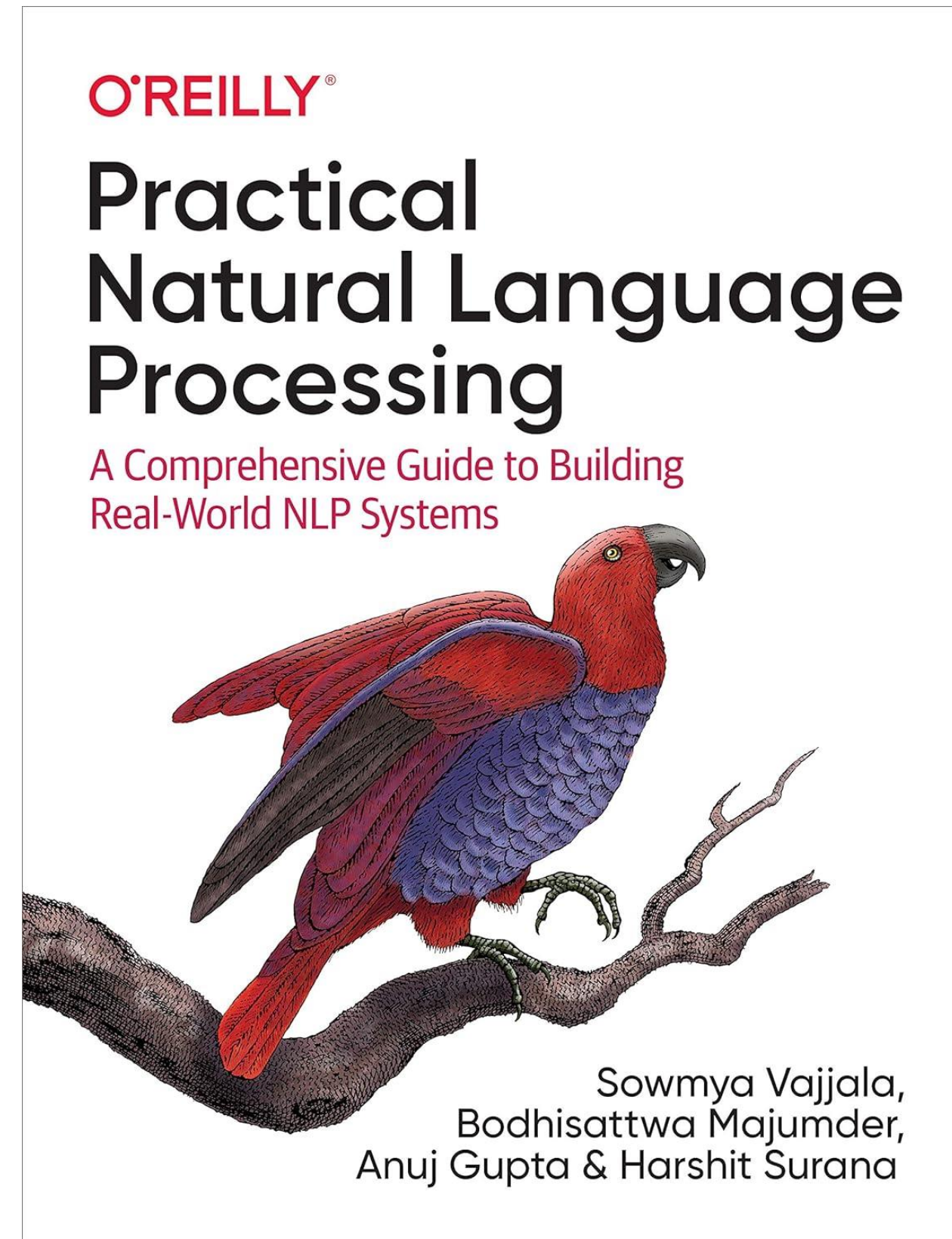
An Introduction to Natural Language Processing,
Computational Linguistics, and Speech Recognition
with Language Models

Third Edition draft

Daniel Jurafsky
Stanford University

James H. Martin
University of Colorado at Boulder

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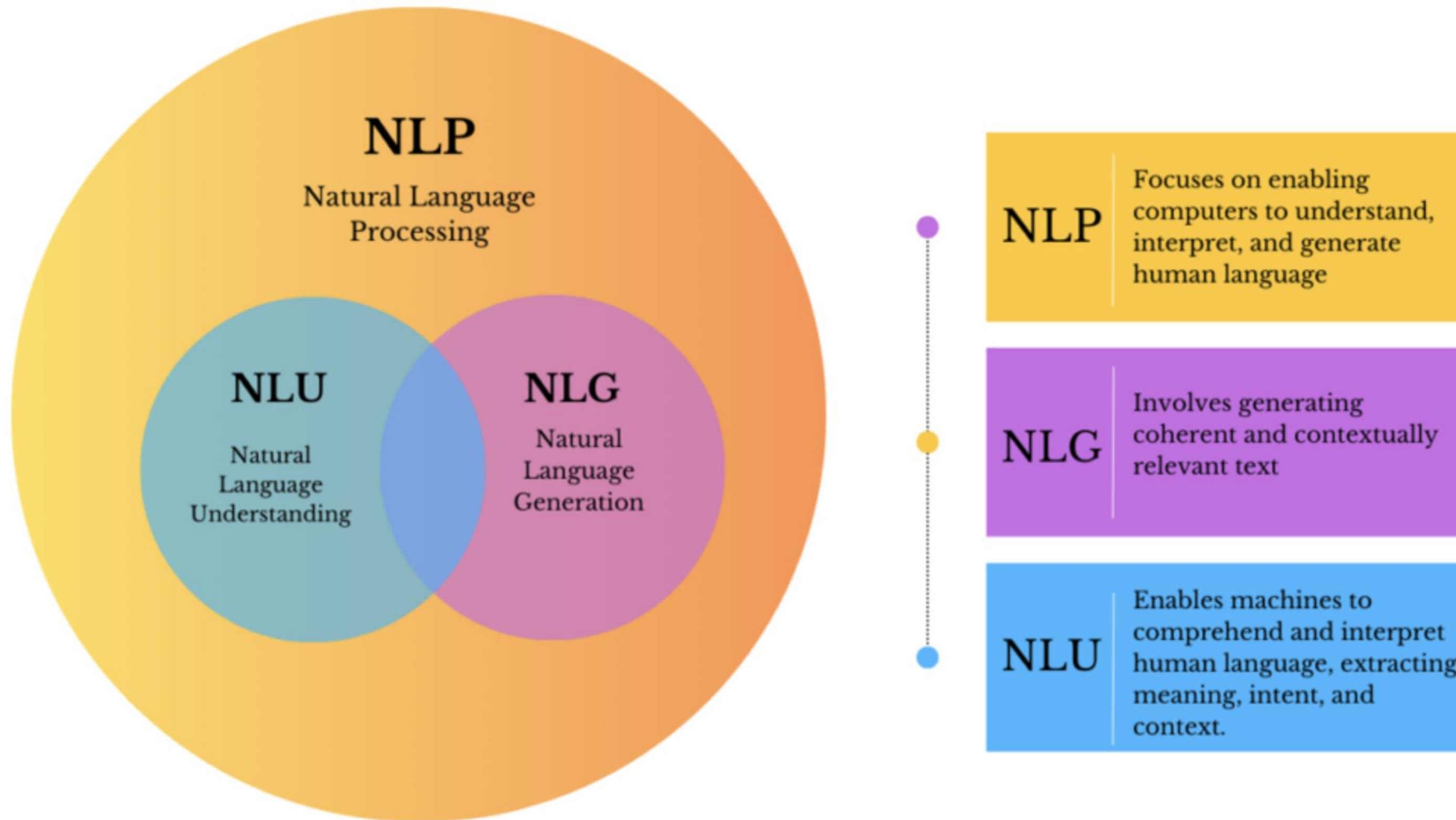


Development Environment Setup

- **1. Install Python (Choose One Option)**
 - **Option 1:** Download & install **Python** from python.org
 - **Option 2:** Install **MiniConda** or **Anaconda (Full)**
 - **Recommendation:** Use **Manual Python** inside **Anaconda** (best balance)
- **2. Create a Virtual Environment**
 - Always create a **venv** for each project
 - Keeps dependencies clean and avoids version conflicts
 - Activate venv → install packages inside it
- **3. Use PIP for Package Management**
 - `pip install package_name`
 - Keep packages updated
 - Manage requirements through requirements.txt
- **4. Deep Learning Frameworks**
 - Install **TensorFlow** or **PyTorch** (prefer PyTorch for research)
 - GPU version optional (if you have NVIDIA GPU)
- **5. Start with Google Colab**
 - Best for beginners
 - Free GPU
 - No installation required
 - Easier than Jupyter Notebook
- **6. Use Proper IDE for Real Projects**
 - **VS Code** (recommended)
 - or **Spyder**
- **7. Build a Kaggle Portfolio**
 - Join competitions
 - Publish notebooks
 - Improve skills + visibility
 - Essential for **Data Scientist** career



Section 1: How it Start



What Is NLP?

- **Natural Language Processing (NLP)**
- Field that enables computers to:
- Process human language (text or speech)
- Understand and analyze language
- Generate and interact using language
- Sits at the intersection of:
 - Computer Science
 - Linguistics
 - Machine Learning / AI



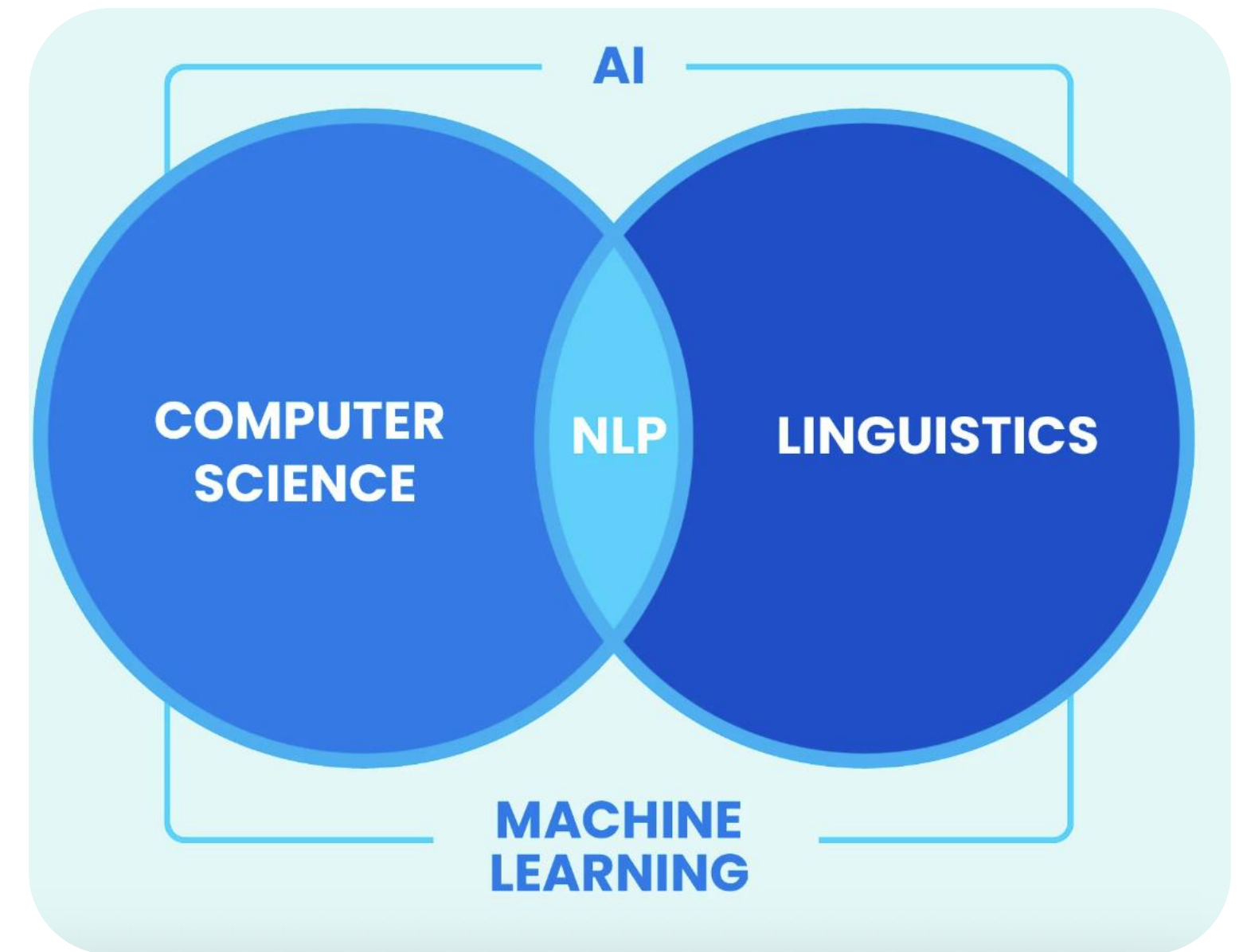
Formal Definition of NLP

- Natural language processing (NLP) is the **processing** of **natural** language **information** by a computer. NLP is a **subfield** of computer science and is closely associated with artificial intelligence. NLP is also related to information **retrieval**, **knowledge representation**, **computational linguistics**, and **linguistics** more broadly. *Eisenstein Jacob*



Linguistics

- Linguistics is the scientific study of language how it is structured, how it is used, and how it is learned.
- Linguistics studies how language is built, including:
 - Phonetics: the physical sounds of speech
 - Phonology: sound patterns in a language
 - Morphology : how words are formed
 - Syntax: sentence structure
 - Semantics: the meaning of words and sentences
 - Pragmatics: how meaning changes with context



How Linguistics is Connected to NLP

- NLP = Computer + Linguistics
- NLP systems need to understand:
- Sounds
- Words
- Grammar
- Meaning
- Context
- Human communication patterns
- All of these come directly from linguistics.



1. Phonetics & Phonology → Speech Recognition / Speech Synthesis

- Linguistics teaches:
 - How sounds are produced
 - How languages structure sounds
- NLP uses this for:
 - ASR: converting speech to text (Google Assistant, Siri)
 - TTS: text-to-speech (e.g., Urdu TTS)
 - Building acoustic models, pronunciation dictionaries, etc.



2. Morphology → Tokenization, Lemmatization, Stemming

- Linguistics explains:
 - How words are formed
 - Roots, prefixes, suffixes
 - NLP uses it for:
 - Breaking text into units
 - Understanding word variations
 - Building morphological analyzers (important for Urdu, Arabic)
 - Example - کتاب → کتابیں → کتابوں
 - NLP must understand these are the *same*
- root*, which comes from **morphology**.

3. Syntax → Parsing & Grammar Models

- Linguistics provides rules for:
 - Sentence structure
 - Word order
- NLP uses this for:
 - Dependency parsing
 - POS tagging
 - Grammar correction
 - Machine translation
- Example:
- **Urdu SOV** structure vs **English SVO**
- This difference is pure linguistics → NLP must model it.

4. Semantics → Meaning Understanding

- Semantics explains:

- Word meaning
- Sentence meaning
- Ambiguity

"Apple" (fruit) vs "Apple" (company)

NLP handles this using semantics.

- NLP uses it for:

- Text classification
- Question answering
- Chatbots
- Named entity recognition

- Example:

5. Pragmatics → Context Understanding / Chatbots

- Pragmatics teaches:
 - How meaning changes with context
 - Implicit meaning
 - Social meaning
- NLP uses this for:
 - Dialogue systems
 - Emotion detection
 - Sarcasm detection
 - Context-aware responses



6. Computational Linguistics → Core of NLP

- This is the branch where **linguistics and computer science meet**.
- How to model languages mathematically
- Algorithms for parsing
- Statistical language models
- Deep learning for language
- This is the direct foundation of modern NLP (transformers, BERT, GPT, LLMs).
- **In One Sentence**
- Linguistics gives the rules of language.
- NLP converts those rules into algorithms so computers can process language.



Example: What Does an NLP System Do?

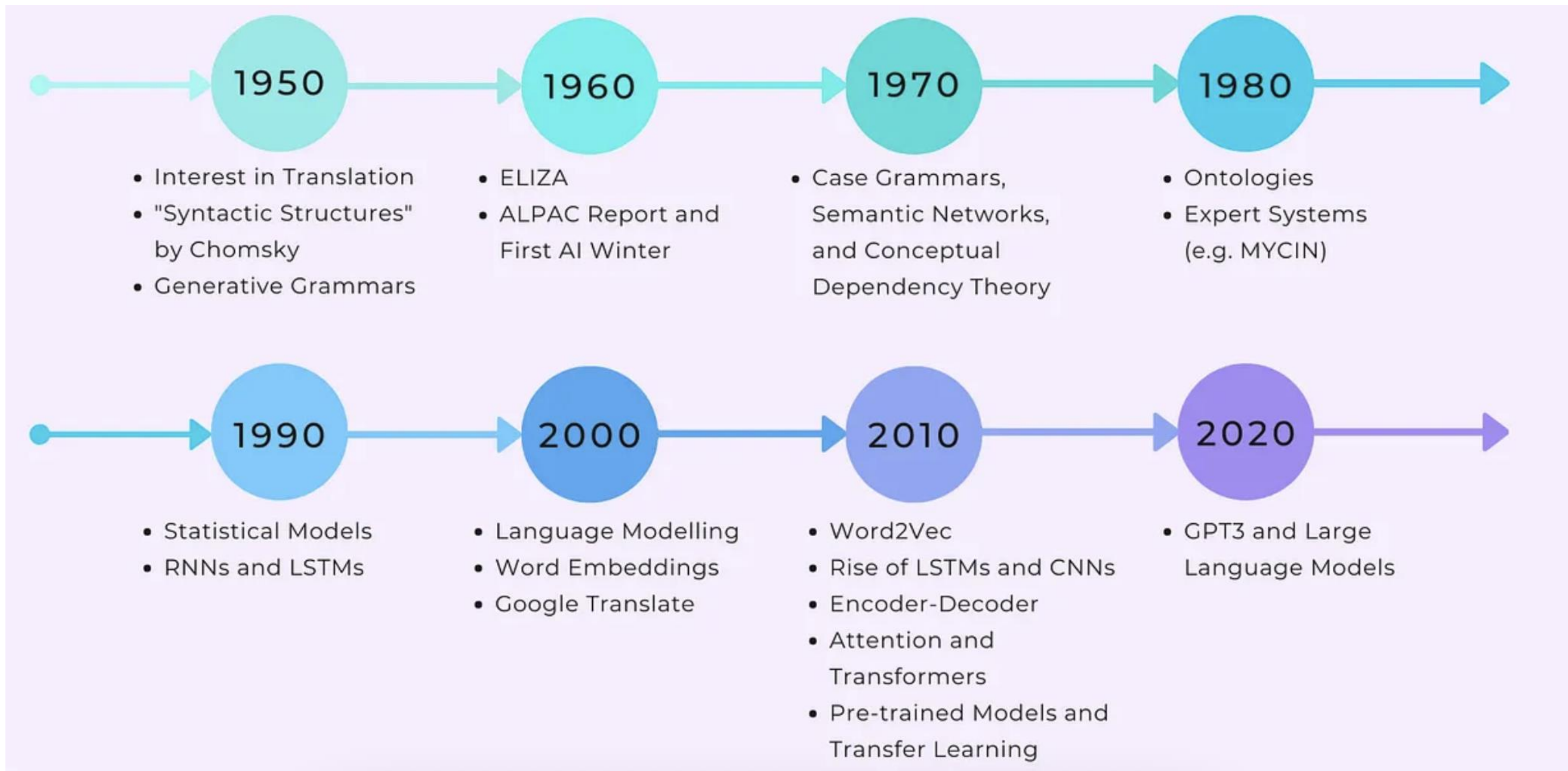
- **User input:**
 - “Book me a flight from Islamabad to Lahore tomorrow morning.”
- **Typical NLP outputs:**
 - Intent: book_flight
 - Origin: Islamabad
 - Destination: Lahore
 - Date: tomorrow
 - Time: morning
- The system then calls downstream services (flight search, booking, etc.)

History of NLP (Rule-Based → Statistical)

- **Early era: Rule-based systems (1960s–1980s)**
- Hand-written grammar rules and patterns
- Example: ELIZA (1966): simple pattern-matching chatbot
- ELIZA is an early NLP computer program developed from 1964 to 1967 at MIT by Joseph Weizenbaum.
- **Statistical NLP (1990s–2000s)**
- Probabilistic models learned from corpora
- N-gram language models, Hidden Markov Model HMM taggers, early MT and ASR
- Strong focus on data, probabilities, and evaluation



History of NLP (Rule-Based → Statistical)



History of NLP (Neural Nets → Transformers → LLMs)

- **Neural NLP (2010s)**
 - Word embeddings (vector semantics)
 - RNNs, LSTMs, CNNs for sequence modeling
- **Transformers (2017)**
 - Attention-based architectures
 - Encoder, decoder, encoder–decoder models
- **Large Language Models (LLMs)**
 - Pretrained on huge corpora by predicting next tokens
 - Instruction tuning and alignment for chat-style assistants



Why NLP Is Important

- **Search & Information Access**
 - Web search, academic search, code search
- **Assistants & Chatbots**
 - Virtual assistants, customer service, tutoring systems
- **Social Media & Opinion Mining**
 - Sentiment analysis, toxicity detection, trend tracking
- **Translation & Multilingual Access**
 - Breaking language barriers
- **Productivity & Creativity**
 - Summarization, drafting, coding assistants, content generation



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Section 2: NLP Applications & Task Taxonomy



Text classification

- **Text classification**
 - Input: sentence, paragraph, document, or message
 - Output: **single label** from a fixed set
- Typical examples:
 - Sentiment: positive / negative / neutral
 - Spam detection: spam / not spam
 - Topic classification: sports, politics, tech, etc.
 - Intent classification: book_flight, cancel_order, get_weather, ...
- **Intent classification** identifies a user's goal or purpose from their text/speech, categorizing it into predefined actions like **"check order status," "book appointment,"** or **"request refund,"**.
- To route it to the correct system response, crucial for **chatbots** and **virtual assistants** to understand and act on user needs efficiently.

Text Classification: Examples

- **Sentiment analysis:**
 - Text: “The food was awful and ridiculously overpriced.”
 - Label: negative
- **Spam detection:**
 - Text: “You won 1,000,000 USD! Click here now!!!”
 - Label: spam
- **Movie review classification using logistic regression:**
 - Features could include word counts or embeddings
 - Model outputs probability of positive vs negative



Sequence Labelling: Definition

- **Sequence labelling**

- Input: sequence of tokens $w_1, w_2, \dots, w_n$
- Output: sequence of labels $y_1, y_2, \dots, y_n$ (same length)

- **Classic tasks:**

- Part-of-Speech (POS) tagging
- Named Entity Recognition (NER)
- Slot filling in dialogue systems

- **Models:**

- HMMs (Hidden Markov Models) and CRFs (Conditional Random Fields), RNNs, Transformers with tagging heads



Sequence Labelling: POS Tagging Example

- **Sentence:**
 - “Marie Curie was born in Warsaw.”
- **POS tags (simplified):**
- Marie / PROPN
- Curie / PROPN
- was / AUX
- born / VERB
- in / ADP
- Warsaw / PROPN
- Use:
 - Helps parsing and understanding syntactic structure

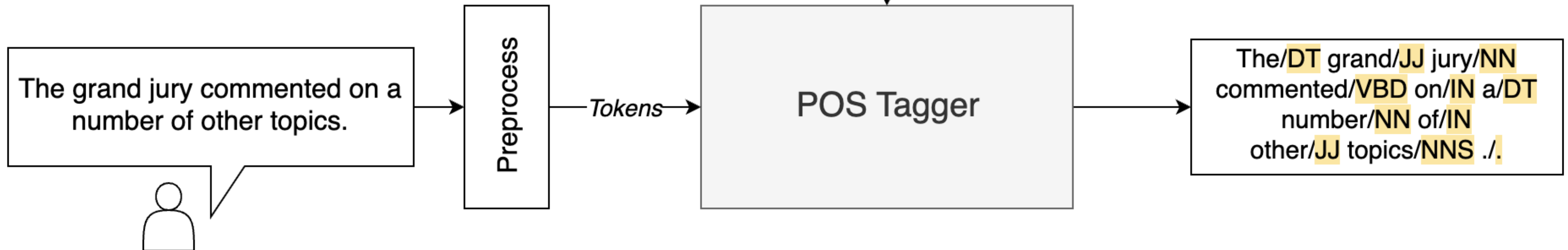
POS Tagging

Tag	Description	Example	Tag	Description	Example	Tag	Description	Example
CC	coordinating conjunction	<i>and, but, or</i>	PDT	predeterminer	<i>all, both</i>	VBP	verb non-3sg present	<i>eat</i>
CD	cardinal number	<i>one, two</i>	POS	possessive ending	<i>'s</i>	VBZ	verb 3sg pres	<i>eats</i>
DT	determiner	<i>a, the</i>	PRP	personal pronoun	<i>I, you, he</i>	WDT	wh-determ.	<i>which, that</i>
EX	existential 'there'	<i>there</i>	PRP\$	possess. pronoun	<i>your, one's</i>	WP	wh-pronoun	<i>what, who</i>
FW	foreign word	<i>mea culpa</i>	RB	adverb	<i>quickly</i>	WP\$	wh-possess.	<i>whose</i>
IN	preposition/ subordin-conj	<i>of, in, by</i>	RBR	comparative adverb	<i>faster</i>	WRB	wh-adverb	<i>how, where</i>
JJ	adjective	<i>yellow</i>	RBS	superlatv. adverb	<i>fastest</i>	\$	dollar sign	<i>\$</i>
JJR	comparative adj	<i>bigger</i>	RP	particle	<i>up, off</i>	#	pound sign	<i>#</i>
JJS	superlative adj	<i>wildest</i>	SYM	symbol	<i>+, %, &</i>	“	left quote	<i>‘ or “</i>
LS	list item marker	<i>1, 2, One</i>	TO	“to”	<i>to</i>	”	right quote	<i>’ or ”</i>
MD	modal	<i>can, should</i>	UH	interjection	<i>ah, oops</i>	(	left paren	<i>[, (, {, <</i>
NN	sing or mass noun	<i>llama</i>	VB	verb base form	<i>eat</i>	)	right paren	<i>],), }, ></i>
NNS	noun, plural	<i>llamas</i>	VBD	verb past tense	<i>ate</i>	,	comma	<i>,</i>
NNP	proper noun, sing.	<i>IBM</i>	VBG	verb gerund	<i>eating</i>	.	sent-end punc	<i>. ! ?</i>
NNPS	proper noun, plu.	<i>Carolinas</i>	VBN	verb past part.	<i>eaten</i>	:	sent-mid punc	<i>: ; ... - -</i>



Sequence Labelling: Example

CC CD DT EX FW IN JJ JJR JJS LS MD NN
NNP NNPS NNS PDT POS PRP PRP\$ RB RBR
RBS RP SYM TO UH VB VBD VBG VBN VBP
VBZ WDT WP WP\$ WRB \$ # " ' () , . :
-----*Tagset*



Sequence Labeling: NER Example

- **Sentence:**
 - “Marie Curie was born in Warsaw.”
- BIO style tagging (Beginning, Inside, Outside) - NER labels:
 - Marie / B-PER
 - Curie / I-PER
 - was / O
 - born / O
 - in / O
 - Warsaw / B-LOC
- **Explanation**
 - **Marie** → start of a person → B-PER
 - **Curie** → continuation of the same person → I-PER
 - **Warsaw** → start of a location → B-LOC
 - All other words are O (not part of any named entity)
 - **Use:** Downstream tasks: information extraction, QA, knowledge base population

Mixed BIO Tagging Activity

- **Task:** Tag the following sentence using BIO format (***B-PER, I-PER, B-LOC, B-ORG, B-DATE, O, etc.***). Some entities have multiple words, so pay attention!
- **Sentence:**
- “Malala Yousafzai visited New York University in the United States on 12 October 2014.”
- Identify and tag the following entity types:
- PER → Person
- LOC → Location
- ORG → Organization
- DATE → Dates
- O → Outside any entity

Token	Tag
Malala	B-PER
Yousafzai	I-PER
visited	O
New	B-ORG
York	I-ORG
University	I-ORG
in	O
the	B-LOC
United	I-LOC
States	I-LOC
on	O
12	B-DATE
October	I-DATE
2014	I-DATE
.	O



Seq2Seq: Definition and Intuition

- **Sequence-to-sequence (seq2seq) tasks**
 - **Input:** text sequence in some format
 - **Output:** new text sequence (potentially different length)
- **Main applications:**
 - Machine translation
 - Summarization
 - Dialogue response generation
 - Paraphrasing and style transfer
- **Typical architectures:**
 - Encoder–decoder with attention / Transformers

Seq2Seq: Definition and Intuition

- **Seq2Seq: Examples**
- **Machine translation:**
- **Input:**
 - “你好，我叫佐海，我住在巴基斯坦
 - Nǐ hǎo, wǒ jiào zuǒ hǎi, wǒ zhù zài bājīstǎn.
- **Output:** Hello, My Name is Zoh Hai. I live in Pakistan.”
- **Summarization:**
 - Input: 5–10 paragraph news article
 - Output: 2–3 sentence summary
- **Dialogue:**
 - Input: conversation context
 - Output: next system **utterance**
- An **utterance** is a single piece of spoken or written text produced by one speaker at one time.
- **Examples of Utterances**
- **User Utterances:**
 - “Hello”
 - “Book a flight for me”
- **System Utterances:**
 - “Hi! How can I help you?”
 - “Sure, where do you want to fly?”
- Each line is one utterance.



Retrieval & Question Answering: Overview

- **Retrieval**

- Given a query, return relevant documents or passages
- Classic information retrieval, vector space models, dense retrieval

- **Question Answering (QA)**

- Given a question (and possibly documents), produce an answer
- QA, reading comprehension, open-domain QA

- **RAG (Retrieval-Augmented Generation)**

- Retrieve relevant passages
- Feed them into an LLM to generate a grounded answer



Retrieval & QA: Example

- **Question:**
 - “When did Beyoncé start becoming popular?”
- **Pipeline:**
 - Retrieve Wikipedia passage about Beyoncé
 - QA model extracts answer: “in the late 1990s”
- **Motivation:**
 - Better factual accuracy than pure generation without retrieval



Multimodal NLP

- **Text + Image**

- Image captioning (generate text given image)
- Text–image retrieval (retrieve image given caption, CLIP-style)

- **Text + Speech**

- ASR (Automatic Speech Recognition) → convert speech to text
- Text-to-Speech (TTS) → generate speech from text
- Spoken dialogue systems (ASR + NLP + TTS)

- **Other modalities**

- Text + code, text + audio, text + video, etc.

Section 3: Datasets & Annotation



Why Do Datasets Matter?

- **Modern NLP is driven by data**
 - Supervised learning requires labeled examples
- **Datasets encode:**
 - Domains, genres, registers
 - Languages, dialects, social groups
- **Good datasets come with:**
 - Documentation (datasheets / data statements)
 - Clear train / dev / test splits
 - Annotation guidelines



Example Datasets

- **Text classification / sentiment**
 - IMDB movie reviews (positive vs negative labels)
 - <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>
- **Sequence labeling**
 - CoNLL-2003 (English NER)
 - <https://www.kaggle.com/datasets/priyanang3003/ner-conll2003>
- **Question answering**
 - Stanford Question Answering Dataset - SQuAD (Wikipedia passages + questions + answer spans)
 - <https://www.kaggle.com/datasets/stanfordu/stanford-question-answering-dataset>
- **Speech**
 - LibriSpeech (ASR; read audiobooks)
 - <https://www.kaggle.com/datasets/renusankhla/librispeech-dataset>
 - Common Voice (crowd-sourced multilingual speech)
 - <https://www.kaggle.com/datasets/mozillaorg/common-voice>



Annotation: What and How?

- **Annotation** = adding labels/structure to raw data
 - POS tags, entities, sentiment, question–answer spans, etc.
- **Annotation requires:**
 - Clear label definitions
 - Detailed examples and counterexamples
 - Procedures for difficult/ambiguous cases
- **Often done by:**
 - Crowdworkers with instructions
 - Trained annotators (e.g., linguists, domain experts)



Annotation Guidelines

- **Annotation guidelines typically include:**
- 1. Task definition and goals
 - A clear description of what the annotation task is and why it is being done.
- 2. Label inventory and definitions
 - A list of all labels (e.g., PER, LOC, ORG) and what each label means.
- 3. Examples of typical cases
 - Sample sentences showing how to apply labels correctly in normal situations.
- 4. Edge cases and how to resolve them
 - Instructions for ambiguous or tricky cases, so annotators make consistent decisions.
- 5. Formatting and quality standards
 - Rules for how annotations must be written, structured, and checked (e.g., BIO format, spelling, spacing, accuracy checks).
- Make annotations **consistent**, **reproducible**, and **explainable**



Inter-Annotator Agreement (IAA)

- Inter-Annotator Agreement (IAA)
- Different annotators will sometimes disagree
 - Especially on subjective tasks (sentiment, toxicity, sarcasm)
- **Inter-Annotator Agreement (IAA):**
 - Measures how consistently annotators label the same data
 - High agreement → clearer task and better guidelines
- Common concepts:
 - Percent agreement
 - More advanced metrics (e.g., Cohen's kappa) in later lectures

Common Data Problems: Noise

- **Common Data Problems: Noise**
- **Noise** in datasets:
 - Typos, spelling mistakes
 - Off-topic content or spam
 - ASR transcription errors in speech corpora
- **Consequences:**
 - Models learn incorrect patterns
 - Lower performance, especially in downstream tasks
- **Mitigation:**
 - Cleaning, filtering, robust models, data augmentation



Common Data Problems: Imbalance

- **Class imbalance:**
 - Some labels are much more frequent than others
 - **Example:** toxicity detection:
 - a. 95% non-toxic, 5% toxic
- **Problem:**
 - Naive classifier predicting “non-toxic” always → 95% accuracy
 - But completely fails on the minority class
- **Techniques (later in course):**
 - Reweighting, resampling, specialized loss functions, better evaluation metrics

Section 4: Challenges in NLP



Challenges: Ambiguity

- **Lexical ambiguity**

- Same word, multiple meanings
- Example: “bank” (river bank vs financial institution)

- **Syntactic ambiguity**

- Multiple possible parse trees
- Example: “I saw the man with a telescope.”
- **Interpretation:**
- I used a telescope → to see the man
- I saw a man → who had a telescope

- **Semantic and pragmatic ambiguity**

- Meaning depends heavily on context and world knowledge
- Example: “Can you pass the salt?” (request, not question about ability)



Ambiguity: Examples to Think About

- **Sentence:**
 - “Visiting relatives can be boring.”
- **Two readings:**
 - Relatives who visit are boring
 - The activity of visiting relatives is boring
- **NLP systems must:**
 - Represent alternative interpretations
 - Use context (previous sentences, world knowledge) to disambiguate



Bias and Fairness in NLP

- **Sources of bias:**

- Training data (who is represented? how?)
- Annotation decisions and guidelines
- Pretrained components (embeddings, LLMs)

- **Manifestations:**

- Toxicity classifiers over-flagging certain identities or dialects
- Stereotypes in generated text (gender, race, profession)

- **Mitigation directions: (the visible effects or symptoms of a problem)**

- Better dataset documentation (datasheets, data statements)
- Model cards for transparency
- Evaluation across subgroups and varieties



Bias Example: Dialect and Identity

- **Observed issues in practice:**
 - Sentences that simply mention identity terms:
 - a. “Many women in tech face challenges.”
 - Sometimes mis-labeled as toxic or negative
- **Impact:**
 - Potential silencing of marginalized groups
- **Takeaway:**
 - Need critical evaluation of datasets and models, not just accuracy numbers

Hallucination in Large Language Models

- **Hallucination:**
 - Model outputs fluent, confident, but **false** information
- Why it happens:
 - Training objective is next-token prediction, not “truth”
 - Over-generalization from patterns in training data
- Risks:
 - Wrong medical, legal, or safety advice
 - Fabricated references, facts, URLs



Mitigating Hallucination (High-Level)

- **Retrieval-Augmented Generation (RAG)**
 - Retrieve relevant documents
 - Have the LLM generate answers conditioned on retrieved text
- **Instruction tuning and alignment**
 - Additional training on curated instruction–response pairs
 - Emphasis on honesty, refusal to answer unsafe questions
- **User interface / system design**
 - Show sources when possible
 - Add warnings, encourage verification



Domain Shift and Robustness

- **Domain shift:**
 - Train and test distributions differ
 - Example:
 - a. Train sentiment model on movie reviews
 - b. Deploy on social media posts or product reviews
- **Effects:**
 - Performance drop, unexpected errors
- **Approaches (later in course):**
 - Domain adaptation, continual learning, robust evaluation on multiple domains



Low-Resource Languages

- The world has thousands of languages
 - Most have very little labeled or even unlabeled data
- Challenges:
 - Lack of corpora, dictionaries, tools, funding, standard orthographies
- **Corpora**
 - Large collections of real text (sentences, documents, conversations)
 - Used to train models, analyze language patterns, frequencies, syntax, etc.
 - Example: A dataset of 1 million tweets, Think: lots of sentences.
- **Dictionaries**
 - A list of words with meanings, pronunciations, or features.
- Used for lookup, spell-checking, morphological info, or word definitions.
- Example: A list of English words with their meanings.
- **Orthography** = the way a language is written.
 - **It includes:** The alphabet (letters you use), The spelling rules, How words are written correctly
- **Consequences:**
 - NLP benefits concentrated on a small set of high-resource languages
- **Active research:**
 - Cross-lingual transfer, multilingual LMs, community-driven data collection



Code-Switching & Language Variation

- **Code-switching:**
 - Mixing languages within a sentence or discourse
 - Very common in informal communication
- **Examples:**
 - “Por primera vez veo a @username actually being hateful! it was beautiful :)”
 - “dost tha or rahega ... don’t worry ... sabar rakhe”
- **Challenges:**
 - Tokenization and tagging become harder
 - Language identification, ASR, MT, and LMs often underperform



Language Is Situated

- Language varies by:
 - Language and dialect (e.g., AAE, regional varieties)
 - Genre (news, social media, conversation, legal text, medical notes)
 - Demographics (age, gender, socioeconomic background)
 - Time (historical change)
- Implication:
 - Models and datasets must be interpreted in context
 - Corpus documentation (datasheets, data statements) is essential



Summary of Today's Lecture

- **NLP:**

- Computational processing of human language (text and speech)

- **Task taxonomy:**

- Text classification, sequence labeling, seq2seq, retrieval & QA, multimodal tasks

- **Data and annotation:**

- Key datasets (IMDB, CoNLL-2003, SQuAD, LibriSpeech, Common Voice)
- Importance of guidelines, IAA, awareness of noise, bias, imbalance

- **Challenges:**

- Ambiguity, bias & fairness, hallucination in LLMs
- Domain shift, low-resource languages, and code-switching



Looking Ahead

- **Next topics in the course:**

- Words and tokens (Unicode, tokenization, subwords, corpora)
- Simple language models (n-gram LMs, perplexity)
- Logistic regression for text classification

- **Recommended reading:**

- Jurafsky & Martin, *Speech and Language Processing*, 3rd ed.
 - a. Introduction
 - b. Words and Tokens
 - c. N-gram Language Models



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Your Tasks



Lab Task (1)

- **Setup and warm-up:**

- Create Colab notebooks / environment.
- Install `torch`, `transformers`, `datasets`, `scikit-learn`, `torchaudio`.

- **Dataset exploration:**

- **Load** and inspect samples from IMDB, CoNLL-2003, SQuAD, and a small speech dataset (e.g., subset of LibriSpeech or Common Voice transcripts).
- **Print statistics:** label distribution, sentence lengths, vocabulary size (approximate).

I will check in next class and assign Marks

